What the guard bought

A bit-for-bit counterfactual of assertion-guarded refitting
in the block-1 positive control

Technical note

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Abstract

The block-1 positive control study (`code/analysis/time_block_forecast/block1_positive_control/`) fits 40 chains — settings $\{4, 5\}$ × methods {i.i.d., AR(2)} × datasets 1–10 — under an assertion guard: a fit that fails truth recovery (B) or predictive-spread (C) is refitted with a new seed, at most three times. The guard replaced the attempt-0 chain in 13 of the 40 cells. This note quantifies exactly what that intervention did to the forecast scores, by refitting every replaced attempt-0 chain bit-for-bit (same seed, same call) and scoring it — i.e. by reconstructing the study an *unguarded* pipeline would have delivered. On the affected cells the guard improved the Brier score by 16% and the CRPS by 12% on average; the rescues are largest exactly where the reflected-ridge law $\max_h \mathrm{sd}(\hat{y}_h) \approx 1.5\,|\hat{\lambda}|$ predicts (up to 12× on Brier for $|\hat{\lambda}| \geq 5$), and negligible for mild reflections. One cell moved the other way: a $\hat{\lambda} = -15$ reflected chain *outscored* its healthy replacement. We argue this exception is not a defect but the point: the guard selects fits that are consistent with the model and the known truth — a criterion computable *before* seeing validation outcomes — and therefore it is not, and must not be, a score optimizer.

1 Setting

The positive control asks whether fitting AR(2)-generated data with the AR(2) model beats an i.i.d.-in-time skew-t baseline in forecast scores. Design: data settings 4 (strong, $\phi = (0.80, -0.35)$, spectral radius 0.592) and 5 (near-unit-root, $\phi = (0.15, 0.80)$, radius 0.973); datasets 1–10; block 1 only (train $t = 1..50$, forecast $t = 51..65$); both methods. Truth: $\lambda = 3$, $\beta_0 = 10$.

Block 1's short prefix is where the reflected ridge (`tex/lambda_phiz_ridge`) bites hardest, so every fit runs the self-consistency assertions of `fit_diag_utils.R` and is kept only if it passes

- **B** truth recovery: $\mathrm{sign}(\hat{\lambda}) = +$, $|\hat{\lambda}/3 - 1| < 0.6$, $\hat{\beta}_0$ near 10;
- **C** predictive spread: $\max_h \mathrm{sd}(\hat{y}_h) \leq 3\times$ marginal SD,

otherwise it is refitted with seed $+\,7919 \cdot$ attempt, at most three reseeds, *symmetrically for both methods* (the ridge is a property of the shared skew-t family: the i.i.d. method reflected five times in this study). A cell still failing after four attempts is kept but flagged, and the primary analysis excludes it. The rule was pre-registered before any scores were computed.

The guard changed the delivered chain in 13 cells: 10 rescued (reseed found a passing chain) and 3 flagged (setting 5: dataset 2 under both methods, and dataset 8 under AR(2)).

2 The counterfactual

`run_block1_guarded.R` saves only the kept fit, so the unguarded study is not on disk — but it is fully reproducible: the attempt-0 seed is `get_tbf_seed(setting, method, dataset)`, and rerunning `mcmc()` with it reproduces the rejected chain bit-for-bit. `guard_effect.R` does exactly this for the 13 affected cells, pushes each chain through `forecast_block()`, and scores both versions under the study protocol: CRPS and the Brier score at the q_{95} threshold of the validation block, averaged over leads 1–5 (the pre-registered primary window). Define

unguarded study = attempt-0 chain in every cell,   guarded study = the kept chains.

The 27 unaffected cells are identical in both, so the whole effect of the guard lives in Table 1.

3 Results

Table 1: All 13 affected cells. m1 = i.i.d., m2 = AR(2); arrows read attempt-0 → kept. "att" = attempts used (4 = flag kept). Scores are means over leads 1–5.

cell		att	$\hat{\lambda}_0$	$\hat{\lambda}_{\text{kept}}$	CRPS		Brier q_{95}	
					att-0	kept	att-0	kept
s4 d5 m2	rescue	2	−7.28	+3.93	2.64	1.57	0.0205	**0.0017**
s4 d5 m1	rescue	2	−7.23	+3.52	3.19	1.90	0.0265	**0.0033**
s4 d10 m1	rescue	2	−4.97	+2.99	2.87	2.21	0.0558	0.0226
s5 d7 m1	rescue	2	−3.99	+2.47	1.61	1.29	0.0832	0.0463
s4 d4 m2	rescue	2	−4.01	+2.19	1.46	1.12	0.0010	0.0022
s4 d6 m2	rescue	3	−8.90	+2.86	5.59	5.48	0.1378	0.1349
s5 d5 m1	rescue	2	−4.28	+2.60	2.80	2.45	0.0164	0.0166
s5 d6 m2	rescue	2	+5.40	+3.95	2.24	2.30	0.0794	0.0799
s5 d8 m1	rescue	4	−2.62	+2.29	1.97	2.37	0.0000	0.0007
s5 d3 m2	rescue	4	**−15.18**	+3.34	1.94	**2.15**	0.1088	**0.1323**
s5 d2 m1	flag	4	+0.23	+0.24	1.17	1.17	0.0194	0.0194
s5 d2 m2	flag	4	+0.55	+0.56	1.16	1.16	0.0190	0.0190
s5 d8 m2	flag	4	−2.47	−2.54	1.23	1.25	0.0000	0.0000

Table 2: Group means over the affected cells (leads 1–5).

group	cells	CRPS		Brier q_{95}	
		att-0	kept	att-0	kept
setting 4, i.i.d.	2	3.03	2.05	0.0412	0.0129
setting 4, AR(2)	3	3.23	2.73	0.0531	0.0463
setting 5, i.i.d.	4	1.89	1.82	0.0298	0.0207
setting 5, AR(2)	4	1.64	1.71	0.0518	0.0578
all affected	13	2.30	2.03	0.0437	0.0368

Three regularities.

1. The rescue size follows the $|\lambda|$ law. The ridge note established $\max_h \mathrm{sd}(\hat{y}_h) \approx 1.49\,|\hat{\lambda}|$ across every fit, healthy or broken. Since an overdispersed, mis-centred predictive distribution is punished by both proper scores, the damage of a reflected chain — and hence the value of replacing it — should grow with $|\hat{\lambda}_0|$. It does. The two $|\hat{\lambda}_0| \approx 7$ reflections (s4 d5) give the largest rescues: Brier 0.0205 → 0.0017 and 0.0265 → 0.0033, i.e. 8–12×; CRPS drops by 40%. The mild $|\hat{\lambda}_0| \approx 2.5$ reflections (s5 d8) score essentially the same as healthy chains — a weak reflection widens the forecast too little to matter. Both methods benefit: the i.i.d. method's two setting-4 reflections were among the worst, and their repair drives that group's Brier from 0.0412 to 0.0129.

2. Overall, honesty was cheap and profitable. Across the 13 cells the guard improved Brier by 16% and CRPS by 12%; folded into the full 40-cell study the guarded version is strictly better calibrated *and* slightly better scoring. The positive-control conclusions (AR(2) beats i.i.d. on CRPS at near-unit-root persistence, −31% at leads 1–5, Wilcoxon $p = 0.040$) were additionally shown to be insensitive to the flag-exclusions: the sensitivity run that keeps all cells gives $p = 0.010$–0.021.

3. One cell went the other way, and it is the most instructive row in the table. In s5 d3 the AR(2) attempt-0 chain sat at $\hat{\lambda} = -15.18$ — the most extreme reflection in the study — yet scored *better* than its healthy replacement (CRPS 1.94 vs 2.15, Brier 0.109 vs 0.132). Dataset 3 is the hardest in setting 5 (both methods score Brier > 0.1 there even when healthy); on a hard validation window, an overdispersed forecast hedges, and a confident, well-calibrated forecast pays for its confidence. A finite window of 15 days can and occasionally will reward the wrong model.

4 Why the exception is the point

Suppose the guard had been built the obvious wrong way: refit until the *score* improves. Then s5 d3 would have kept the $\hat{\lambda} = -15$ chain — a fit whose skew loading has the wrong sign by a factor of five, whose latent path is upside down, and whose predictive spread is inflated 5× — because it happened to win on one 15-day window. That is selection on outcomes: it biases every downstream comparison and cannot be defended.

The actual guard never looks at validation data. Assertions B and C are computable from the fit and the training data alone (B uses the simulation truth, which a simulation study owns by construction; C uses the climatology bound that any stationary forecast must respect). The guard therefore implements *selection on model consistency*: it asks "is this chain a sample from the posterior the model claims?", not "did this chain get lucky?". The price is occasionally visible — about +0.02 Brier on one cell here — and it is the premium for an unbiased study. The correct reading of Table 1 is not "the guard won 9 rows and lost 1" but "the guard delivered 37 of 40 cells with certified-consistent fits, and told us exactly which 3 it could not certify."

The flagged cells carry information too. Dataset 2 of setting 5 failed in the same way under *both* methods and all four seeds: $\hat{\lambda} \approx 0.25$ (i.i.d.) and ≈ 0.56 (AR(2)), stable to three decimals across reseeds. This is not chain capture — it is the *data*: under a near-unit-root z process ($h^* \approx 109$ days), a single 50-day training window can realize an essentially flat z path, in which case λ is genuinely under-identified by that window and no reseed can fix it.

The A′ statistic separates the two situations cleanly in its first field test: passing fits have $\mathrm{sd}_t(\hat{z})/(\hat{\sigma}\sqrt{1-2/\pi}) \approx 1.04$–$1.17$, flagged fits 0.57–0.77. Flagging, rather than silently averaging such cells into the comparison, is the guard doing for the study what assertions A/C did for the `z.init` bug: converting a quiet distortion into a visible, attributable one.

5 Summary

- An unguarded block-1 study would have shipped 10 reflected or inflated chains among 40; the guard replaced them at a cost of $\sim$1 extra fit per affected cell.
- Effect on scores, affected cells: Brier −16%, CRPS −12%; rescue magnitude $\propto |\hat{\lambda}_0|$, up to 12× on Brier, negligible below $|\hat{\lambda}_0| \approx 3$.
- Both methods needed it: the ridge is a family property, not an AR(2) property.
- One cell paid a small score premium for consistency; that premium is the operational difference between guarding on *assertions* and cherry-picking on *outcomes*.
- The three flagged cells are a data-level identifiability statement about short windows under near-unit-root persistence, now visible instead of silent.

  A guard that can never lose points against an oracle is not a guard; it is a leaderboard. The value of assertion-guarded refitting is that its selections are defensible before the validation data are opened.